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Effectiveness of static text adversarial methods on continuously updating models ★

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ABSTRACT

Deep neural networks developed for natural language processing tasks have shown to be vulnerable to attacks, and a variety of text adversarial methods have been proposed so far for the purpose of enhancing the attack success rates and attack efficiency in such natural language processing applications. These methods have good attack success rates and efficiency, and can stably generate successful adversarial samples for a "static" victim model (i.e., a fixed model). In reality, as the training data are continually generated, the models will also be continuously updated. Whether the current adversarial attack methods can maintain their performance for these "dynamic" victim models is an open question. In this paper, we design a new task and investigate the performance of static text adversarial attack methods in continuously updating model scenarios. To standardize this task, we propose a comprehensive evaluation framework consisting of two novel experimental methods, three new metrics, and a lightweight dynamic baseline. This framework explicitly decouples the assessment of dynamic attack performance into direct effectiveness and iterative persistence. Extensive experiments on two victim models on three datasets show that a continuously updating model leads to a significant degradation in the performance of static adversarial attack methods. Moreover, the results demonstrate that our framework precisely characterizes the temporal variations of attack performance, confirming its effectiveness and necessity.

1. Introduction

Deep neural networks (DNNs) have achieved remarkable success in various domains including computer vision (Minaee et al., 2021), natural language processing (Otter et al., 2020),and other real-world application areas (Qian et al., 2025). However, at the same time, these models have been found to be extremely vulnerable to adversarial attacks. Attackers generally add undetectable malicious perturbations to the original input to generate adversarial samples, for the purpose of making the DNN model to mismatch (Shafahi et al., 2023).

Adversarial attacks occur in many real-life scenarios where machine learning models and DNNs are used. For instance, simple disguises like glasses or hats can lead to misidentification in face recognition systems, threatening their reliability and trustworthiness (Sun et al., 2024). In addition, in the medical imaging domain, the inherent structural and textural similarities add complexity to image retrieval, increasing the sensitivity of traditional methods and deep learning-based hashing technique methods to adversarial samples (Ding et al., 2023). Thus, it is of great importance to develop effective adversarial attack methods in order to enhance the robustness of DNNs, ensuring information security and even life safety (Chen et al., 2023).

Adversarial attack methods can be categorized into white-box and black-box attacks according to the visibility of the internal structure of the DNNs. White-box attacks can obtain the internal information of the victim model, such as gradient propagation, model structure, loss function, etc., in addition to the output results (Qian et al., 2023). Thus, compared to black-box attacks, white-box attack methods can generate adversarial samples of higher quality. Black-box attacks can only obtain the outputs of the victim model and acquire nothing about the internal information of the victim model, and thus they often need to access the victim model multiple times (Qi et al., 2022). Since black-box attack methods can only obtain the outputs, they have better generalizability and are more closely aligned with realistic adversarial attack scenarios than white-box attacks (Qian et al., 2024).

In the field of image processing, the continuity of images allows various adversarial attacks that preserve visual appearance while effec-

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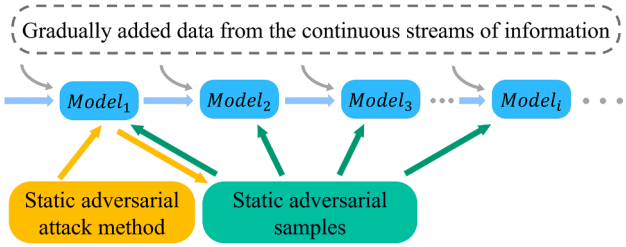


Fig. 1. The process of static text adversarial attack method attacking the continuously updating model. Gray and blue arrows denote iterative model training as the dataset expands; yellow arrows indicate the generation of static adversarial samples at the initial time step; and green arrows represent their attacks on the model across each time step.

tively deceiving DNNs. In the field of NLP, because of the discreteness of texts, NLP models are more vulnerable than image models, while also making adversarial sample generation more challenging due to the risk of producing unnatural samples. For example, character-level attacks are prone to produce wrong words (Huang et al., 2022), while sentence-level attacks are easy to produce semantic distortions (Qi et al., 2021). An ideal adversarial sample should ensure both semantic similarity and syntactic correctness while confusing the victim model. Therefore, many researchers have turned to word-level attacks (Li et al., 2021), which have achieved higher attack success rates and better attack efficiency. Word-level attacks create adversarial samples by adding, replacing, or removing words/phrases with semantically and lexically similar alternatives, ensuring contextual coherence.

However, the above text adversarial attack tasks are all based on the assumption that the amount of dataset for training the DNNs is fixed and, thus, victim models are fixed (Mundt et al., 2023). In reality, the dataset for a given task always expands, and the model needs to be trained on the additional data to ensure good model performance (De Lange et al., 2021). In these circumstances, the model would be trained iteratively to become increasingly robust and adaptable (Wang et al., 2024). Since the current model is modified from the previous model, we call this model sequence as "continuously updating models" in this paper. Existing research on textual adversarial studies are limited to fixed model scenario and without considering this more realistic continuously updating model scenario. Hence, it is of great significance to make text adversarial methods more adaptive to continuously updating models so as to improve the adaptability of AI systems to reality.

There is emerging research combining incremental or continual learning with adversarial attacks, but these studies typically adopt a model-centric perspective. They generally focus on either 1) enhancing the model's adversarial robustness throughout the continual learning process (Bai et al., 2023), or 2) utilizing adversarial mechanisms to mitigate catastrophic forgetting (Kumari et al., 2022). In contrast, this work presents a distinct sample-centric perspective. Unlike existing studies that prioritize optimization of model robustness, we systematically investigate the temporal validity of adversarial samples.

Specifically, it is crucial to study how the effectiveness of existing text adversarial attacks changes as training data grows and models continuously update. Meanwhile, although the most current black-box word-level text adversarial methods have achieved good performance in both attack efficiency and adversarial sample quality, the generation of adversarial samples is still a very time-consuming process (Du et al., 2021). Some adversarial methods are even more costly than a DNN training process itself. Therefore, it would be unimaginably expensive to perform text adversarial attacks continuously when it is already computationally expensive to iterate the continuously updating model. Thus, it is obvious that only the static adversarial samples generated by attacking the original victim model can be utilized to attack the remaining continuously updating models, just as Fig. 1 shows.

Based on the above situation and assumptions, this paper investigates the effectiveness of static text adversarial methods in continuously updating models and introduces a new task. However, traditional static evaluation protocols ignore the temporal dependency of adversarial attacks on this dynamic setting. Therefore, to standardize research in this area, we further propose a comprehensive evaluation framework. The main contributions are summarized as follows:

(1) We introduce the task of "static text adversarial attacks in continuously updating model scenarios", which adopts a distinct sample-centric perspective to investigate the temporal validity of static adversarial methods against continuously updating models.
(2) We propose a comprehensive evaluation framework consisting of novel experimental methods, evaluation metrics, and a lightweight dynamic baseline. This framework explicitly decouples the assessment of dynamic attack performance into direct effectiveness and iterative persistence, while the baseline provides a dynamic reference for benchmarking static methods.
(3) Experimental results show that the effectiveness of static text adversarial methods is significantly affected in these scenarios. The proposed experimental methods and metrics effectively capture the temporal variations of attack effectiveness, while the dynamic baseline demonstrates superior adaptability, verifying the validity and necessity of our evaluation framework. We also provide the suite of datasets and models employed in our experiments to support further research.[1]

The remainder of the paper is organized as follows. Section 2 defines the new task and formulates the special terminology used in this paper. Section 3 describes the proposed evaluation framework. Section 4 presents the experimental results and analysis. Section 5 concludes the paper.

2. Preliminaries

Before exploring the effectiveness of static text adversarial methods on continuously updating models, this section first defines the related terms involved in this paper.

2.1. Definition of continuously updating model and dynamic attack

Continual learning ability is essential for computational systems and autonomous agents that interact with the real world and process continuous streams of information (Parisi et al., 2019). However, this remains a long-term challenge for machine learning and neural network models (Hassabis et al., 2017), for the reason that DNNs can only typically learn representations from fixed batches of training data, without usually considering that the availability of information increases over time. Therefore, to maintain good performance, the model needs to be trained continuously based on the current model as new data is added (Paleyes et al., 2022). Since the model updates constantly, we define it as "*continuously updating model*".

Given an initial model $\mathcal{M}_1$, as the model updates dynamically over time steps $1, 2, \ldots, i, \ldots$, we can get a continuously updating model sequence $\mathcal{M} = \mathcal{M}_1, \mathcal{M}_2, \ldots, \mathcal{M}_i, \ldots$ as illustrated in Fig. 2. Here we denote $\mathcal{M}_i$ as the model at time step i. Thus, the adversarial attack is also conducted over the time steps and is called *dynamic attack*.

Additionally, research on adversarial sample transferability has shown that samples crafted for one model can often mislead others (Yuan et al., 2021). However, current studies on transferability primarily focus on the transfer performance between different static models, without considering how the effectiveness of adversarial samples changes as the model itself evolves through continual learning. Therefore, this work focuses on the behavior of adversarial samples under

[1] Available at https://github.com/hahahage/static_adv_CUmodels

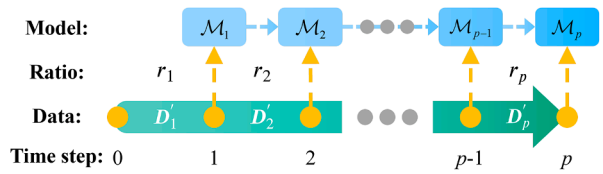


Fig. 2. Illustration of the continuously updating model training in the experiment. The green arrow represents the dataset, which is split into p parts by time steps marked with yellow dots. The blue boxes show the state of the continuously updating model at each corresponding time step.

continuously updating models, expanding the current perspective of adversarial attack research.

With regard to intersection of adversarial attacks and incremental or continual learning, previous studies typically adopt a model-centric perspective. They integrate adversarial samples into the training process to influence model optimization, aiming to enhance robustness or mitigate catastrophic forgetting. Conversely, this work specifically considers scenarios where the model updates are driven solely by non-adversarial incoming data. We focus on the temporal validity of adversarial samples and systematically investigate the impact of model updates on their performance.

Note that while the continuously updating model can be considered "dynamic", we avoid defining it as a "dynamic model" since this term already carries a specific definition in other fields.

2.2. Static word-level adversarial attacks

In text categorization tasks, DNNs should predict the correct label for a particular input sentence. A text adversarial method generates an adversarial sample (i.e., sentence) by adding perturbations to the corresponding original input sample in the dataset, which in turn gives a set of adversarial samples $\boldsymbol{X}'$. Note that the original sample, to which the perturbations are added, should be predicted correctly by the victim model. An ideal adversarial sample should satisfy two requirements: 1) For DNNs, it can successfully trick the model into predicting it as another label, different from the real one. 2) For humans, it can be perceived by humans as still belonging to the real label, with no obvious grammatical or other errors. The samples that meet these requirements in $\boldsymbol{X}'$ are called the valid samples, and the set they form is called the valid sample set $\boldsymbol{X}^{\text{Valid}} \subseteq \boldsymbol{X}'$.

For word-level adversarial methods, the basic method of adding perturbations is to replace words directly. Recently, many perturbation methods that add or remove words have been proposed. These methods include those of splitting phrases, merging phrases, etc. Essentially, word-level text adversarial attack is a combinatorial optimization problem (Nemhauser & Wolsey, 1988), which can be divided into two stages: reduce the search space stage, and the search stage. The reduction of the search space aims to eliminate invalid and low-quality potential adversarial samples. Candidate replacement methods, including those of word embedding-based methods (Sato et al., 2018), language model-based ones (Zhang et al., 2019), synonym-based ones (Ren et al., 2019) and sememe-based ones (Zang et al., 2020), prepare candidate words for each word in the original sample. During the search stage, an optimization algorithm, such as gradient descent (Papernot et al., 2016) and genetic algorithms (Alzantot et al., 2018), is employed to find valid adversarial samples in the reduced search space. More recently, masked language models (MLMs) have been applied in the above two stages due to their excellent contextual comprehension and extensive pre-trained knowledge, with sound results achieved.

Although many text adversarial methods achieve high attack success rates currently, they generally incur high costs. Some methods are time-consuming since they need a great number of queries (Hu et al., 2024), and some take up a large amount of memory or graphics memory space. Because these methods struggle to balance success rate and attack costs, it is difficult to generate new adversarial samples by following the change of the model that is continuously trained and updated with new data. Therefore, in this paper, we define the existing text adversarial methods as "*static*" methods. Thus, we define the new task as *static text adversarial attacks in continuously updating model scenarios*.

3. The proposed evaluation framework

Due to constantly changes of the victim continuously updating model, the text adversarial method has to perform multiple attacks. However, the existing experimental methods and evaluation metrics are for fixed victim models, not suitable for evaluating adversarial methods in the scenarios of continuously updating model. Therefore, in this paper, we propose a new evaluation framework for this new task, which consists of novel experimental methods and evaluation metrics to standardize the conduct of experiments, along with a lightweight dynamic attack baseline introduced to facilitate a direct comparison with static adversarial methods.

3.1. Experimental methods for dynamic attacks

In this paper, we use the method in Fig. 2 to obtain a continuously updating model sequence. First, in the data processing stage, the entire dataset $\boldsymbol{D}$ is split into p parts according to a specific ratio $r = r_1 : r_2 : \ldots : r_p$. Each sub-dataset $\boldsymbol{D}'_i, i \in [1, p]$ represents the newly added data for the model between time steps $i-1$ and i. In the experiment, we first shuffle $\boldsymbol{D}$ and then split it. Meanwhile, $\boldsymbol{D}'_i = \left\{\boldsymbol{D}'_{i,train}, \boldsymbol{D}'_{i,val}, \boldsymbol{D}'_{i,test}\right\}$ denotes the training, validation, and test subsets of $\boldsymbol{D}'_i$, each split according to r.

Then, in the model training stage, each current model $\mathcal{M}_i$ is trained on the basis of the previous model $\mathcal{M}_{i-1}$ by using all the data $\boldsymbol{D}_i = \left\{\boldsymbol{D}'_1, \boldsymbol{D}'_2, \ldots, \boldsymbol{D}'_i\right\}$ accumulated up to the current time step i. This is because the model at subsequent time steps can be trained on all the data accumulated previously up to that point. It should be noted that the model $\mathcal{M}_1$ is trained on $\boldsymbol{D}_1 = \boldsymbol{D}'_1$. In this way, a continuously updating model sequence is formed.

Finally, in the attack stage, due to attack cost constraints, adversarial samples cannot be generated immediately after model training. Therefore, we only attack on the first model $\mathcal{M}_1$ to generate the adversarial samples set $\boldsymbol{X}'$. That is, we perform the static text adversarial attack as described in Section 2.2. Let $\boldsymbol{X}^{\text{Valid}}$ be the subset of $\boldsymbol{X}'$ that can perform a valid attack on $\mathcal{M}_1$. Since the adversarial samples are generated on $\mathcal{M}_1$, it is obvious that only valid samples should be retained when attacking the latter model $\mathcal{M}_2$, that is, only the $\boldsymbol{X}^{\text{Valid}}$ should be used to attack. For the subsequent models, we design two experimental methods as no new adversarial samples are generated. They are distinguished according to whether they use only the samples valid on the previous model $\mathcal{M}_{i-1}$ to attack $\mathcal{M}_i$, aiming to assess different capabilities of the adversarial methods. These two methods are described as follows.

(1) **Single Dynamic Attack** consistently uses $\boldsymbol{X}^{\text{Valid}}$ to attack all models $\mathcal{M}_1, \mathcal{M}_2, \ldots, \mathcal{M}_i, \ldots$ directly, targeting the direct effectiveness against individual models.
(2) **Continuous Dynamic Attack** uses only the sample set $\boldsymbol{X}^{\text{Valid}}_{i-1}$, which was valid for the attack on $\mathcal{M}_{i-1}$, to attack $\mathcal{M}_i$. Then, the samples that are valid for the attack on $\mathcal{M}_i$ are used to form a new set of valid samples $\boldsymbol{X}^{\text{Valid}}_i$, where $\boldsymbol{X}^{\text{Valid}}_1 = \boldsymbol{X}^{\text{Valid}}$, thereby targeting the iterative persistence throughout the updating process.

3.2. Evaluation metrics for dynamic attacks

The traditional Attack Success Rate (ASR) is a static metric designed for fixed models, which fails to capture the dynamic characteristics of our scenario. First, ASR treats attacks at different time steps as independent events, thus ignoring that attack effectiveness is time-dependent. Second, it cannot decouple the performance of adversarial methods into

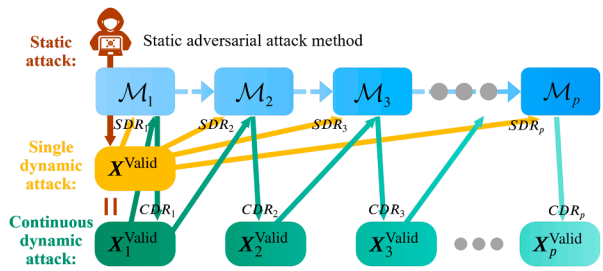


Fig. 3. Overview of the proposed experimental methods and the application of evaluation metrics. The yellow and green arrows represent the attack processes of the proposed single and continuous dynamic attack, respectively, while the reddish-brown arrow illustrates the existing static attack process. Note that $\boldsymbol{X}_1^{\text{Valid}} = \boldsymbol{X}^{\text{Valid}}$.

two distinct capabilities: direct effectiveness and iterative persistence. Specifically, direct effectiveness refers to the ability of samples to successfully attack $\mathcal{M}_i$ directly, regardless of their status at previous time steps, and iterative persistence denotes their ability to maintain validity throughout continuous model updates. Therefore, we propose three novel evaluation metrics as follows.

3.2.1. Single dynamic attack success rate

Single Dynamic attack success Rate SDR_i is used to measure the direct effectiveness of adversarial methods in the single dynamic attack, which is defined by,

$$SDR_i = \left|\{X'_j \mid F_i(X'_j) \text{ is True}, X'_j \in \boldsymbol{X}^{\text{Valid}}\}\right| \Big/ \left|\boldsymbol{X}^{\text{Valid}}\right| \tag{1}$$

where $|\cdot|$ denotes set size, the numerator denotes the number of samples in $\boldsymbol{X}^{\text{Valid}}$ that can confuse $\mathcal{M}_i$, and $F_i(X'_j)$ indicates whether the adversarial sample X'_j succeeded in confusing $\mathcal{M}_i$. If $i = 1$, then $SDR_i = 1.0$.

3.2.2. Continuous dynamic attack success rate

Continuous Dynamic attack success Rate CDR_i is used to measure the iterative persistence of adversarial methods in the continuous dynamic attack, which is computed by,

$$CDR_i = \left|\boldsymbol{X}_i^{\text{Valid}}\right| \Big/ \left|\boldsymbol{X}_1^{\text{Valid}}\right| \tag{2}$$

where $\boldsymbol{X}_i^{\text{Valid}}$ denotes the set of valid samples on the model $\mathcal{M}_i$ which is defined in the previous subsection. Note that $\boldsymbol{X}_1^{\text{Valid}} = \boldsymbol{X}^{\text{Valid}}$. Obviously when $i = 1$ *or* 2, $SDR_i = CDR_i$.

3.2.3. Overall dynamic attack success rate

In addition to SDR_i and CDR_i, we further define the overall dynamic attack success rate (*Overall*). This metric integrates the results of single and continuous dynamic attack to provide a comprehensive evaluation of static adversarial methods in continuously updating model scenarios. It is formulated as,

$$Overall = 2 \cdot \frac{Single \cdot Continuous}{Single + Continuous} \tag{3}$$

where *Single* and *Continuous* represent two intermediate terms reflecting the performance of adversarial methods on the single and continuous dynamic attack, respectively, and are formally defined as,

$$Single = 1 - \frac{\sum_{i=2}^{p}[(SDR_1 - SDR_i)\frac{r_i}{\sum_{k=1}^{i} r_k}]}{p-1} \tag{4}$$

$$Continuous = 1 - \sum_{i=2}^{p}[(CDR_{i-1} - CDR_i)\frac{r_i}{r_{i-1}+r_i}]. \tag{5}$$

In Eq. (4), $(SDR_1 - SDR_i)\frac{r_i}{\sum_{k=1}^{i} r_k}$ reflects the decrease in the efficiency of adversarial samples between $\mathcal{M}_1$ and $\mathcal{M}_i$. This is because single dynamic attack always uses the first set of valid adversarial samples $\boldsymbol{X}_1^{\text{Valid}}$ (i.e., $\boldsymbol{X}^{\text{Valid}}$) against the continuously updating model. Meanwhile, since earlier models differ less from $\mathcal{M}_1$, the resulting reduction in efficiency of the adversarial samples should be less. Therefore, we let the weight $\frac{r_i}{\sum_{k=1}^{i} r_k}$ decrease with i in order to emphasize the importance of the early time steps.

In Eq. (5), $(CDR_{i-1} - CDR_i)\frac{r_i}{r_{i-1}+r_i}$ reflects the efficiency reduction between two neighboring time steps, because unlike single dynamic attack, continuous dynamic attack is an iterative process. After each attack on $\mathcal{M}_i$, the set $\boldsymbol{X}_i^{\text{Valid}}$ for the next attack is updated.

Therefore, we define the *Overall* measure using the harmonic mean of *Single* and *Continuous*, as shown in Eq. (3). The rationale for adopting a harmonic mean-based formulation, rather than a simple arithmetic or geometric mean, lies in its distinct sensitivity to the lower value among the components. In the context of continuously updating models, an ideal adversarial method must possess two concurrent capabilities, that is, direct effectiveness against individual models (captured by *Single*) and iterative persistence throughout the continuous updating process (captured by *Continuous*). A method that excels in one aspect but fails in the other (e.g., showing high direct effectiveness but poor iterative persistence) cannot be considered robust in the scenarios of continuously updating model. Consequently, the harmonic mean enforces a strict trade-off, penalizing the methods that exhibit unbalanced performance. This design establishes the *Overall* metric as a stringent standard within our proposed evaluation framework, ensuring that a high score strictly reflects the comprehensive effectiveness of an adversarial method in both the single and continuous dynamic attack.

Table 1
Summary of the three benchmark datasets.

Dataset	#Classes	Training	Validation	Test	AvgWord
IMDB	2	25,000	0	25,000	234
SST-2	2	6920	872	1821	17
SNLI	3	550,152	10,000	10,000	8

3.3. Overview of experimental methods and evaluation metrics

The attack procedures of the proposed experimental methods and the application of the evaluation metrics are illustrated in Fig. 3. For a continuously updating model $\mathcal{M}$, we first employ the existing static adversarial methods to attack the initial model $\mathcal{M}_1$, which we refer to as the "static attack". Then, the adversarial samples that successfully attack $\mathcal{M}_1$ in the static attack are collected as $\boldsymbol{X}^{\text{Valid}}$ (i.e., $\boldsymbol{X}_1^{\text{Valid}}$).

In the single dynamic attack, $\boldsymbol{X}^{\text{Valid}}$ is directly used to attack $\mathcal{M}_i$, and the direct effectiveness is measured by SDR_i. In contrast, in the continuous dynamic attack, $\boldsymbol{X}_{i-1}^{\text{Valid}}$ is used to attack $\mathcal{M}_i$, with the iterative persistence measured by CDR_i. The adversarial samples that successfully attack $\mathcal{M}_i$ are then selected to form $\boldsymbol{X}_i^{\text{Valid}}$. Finally, *Overall* is calculated by using Eqs. (3)–(5) to evaluate the comprehensive effectiveness of static methods in continuously updating model scenarios.

3.4. Lightweight dynamic attack baseline

To ensure a rigorous evaluation within our proposed framework, it is essential to establish a reasonable dynamic attack baseline to help discern the underlying causes of variations in the effectiveness of static adversarial samples. Given the absence of dedicated dynamic text adversarial attack methods in current research, we construct a lightweight dynamic attack baseline, termed DynHI.

Unlike static methods that generate adversarial samples only once, and distinct from attacks that restart the optimization process from scratch at every time step, DynHI is based on the HIQPSO algorithm (Chen et al., 2023) and employs a "warm-start" re-optimization strategy. Specifically, DynHI retains both the generated adversarial samples

and their corresponding optimization states (i.e., the candidate populations in the particle swarm) from the initial or previous time step. These preserved states allow DynHI to reuse prior search information for rapid re-optimization in subsequent time steps. The specific execution of DynHI under the two proposed experimental methods is as follows:

- **DynHI for Single Dynamic Attack**: For each adversarial sample in $\boldsymbol{X}^{\text{Valid}}$ at $i \geq 2$, DynHI first verifies whether it successfully attacks $\mathcal{M}_i$. If it fails, DynHI does not re-initialize a population; instead, it resumes the HIQPSO algorithm from the population state saved at $i = 1$. If this resumed optimization produces a valid adversarial sample, the attack is considered successful.
- **DynHI for Continuous Dynamic Attack**: At $i \geq 2$, for an adversarial sample in $\boldsymbol{X}_{i-1}^{\text{Valid}}$, DynHI checks its validity against $\mathcal{M}_i$. If the sample fails, DynHI resumes the optimization process using the population state saved at $i-1$. Once successful, the newly generated sample is added to $\boldsymbol{X}_i^{\text{Valid}}$, and its corresponding population state is preserved to support the attack at $i+1$.

4. Experiments

The experiments were conducted for the evaluation of effectiveness of static text adversarial methods under continuously updating models. This section will first describe the datasets, the victim models, and the adversarial attack methods used in the experiments together with other details. After that, the experimental results and analysis are provided.

4.1. Datasets, victim models and adversarial methods

For the datasets, we chose three classical NLP datasets, namely IMDB (Maas et al., 2011), SST-2 (Socher et al., 2013) and SNLI (Bowman et al., 2015). The former two datasets are used for the sentiment classification task, and the latter one is for the natural language inference classification task. Both IMDB and SST-2 are in the domain of movie reviews. They are both bicategory datasets, where the input is a sentence, with the two labels being positive and negative, respectively. But IMDB inputs are longer on average (234 words) than those in SST-2 (17 words). SNLI is a human handwritten dataset. It is a triple category dataset where the inputs are premise-hypothesis sentence pairs, with three labels being entailment, contradiction, and neutral, respectively. Detailed information about the datasets is listed in Table 1. As IMDB lacks a validation set, we split the training set into 80% training and 20% validation.

For the victim continuously updating model, we chose two widely used classical models in the field of NLP, namely, the bi-directional long and short-term memory (BiLSTM Conneau et al., 2017) model and the transformer-based BERT model (Devlin et al., 2019).

For the adversarial methods, we chose seven static word-level text adversarial methods to attack the continuously updating models and produce static adversarial samples. The first method is based on word embedding and uses genetic algorithm (GA) to search for adversarial samples (Alzantot et al., 2018). The second method, called "PWWS" (Ren et al., 2019), searches for adversarial samples in a synonym space using a greedy algorithm. The next three methods construct a sememe-based solution space and use three different particle swarm optimization (PSO) algorithms to search for adversarial samples. The three PSO algorithms are discrete PSO (DPSO) (Zang et al., 2020), discrete quantum-behaved PSO (DQPSO) (Chen et al., 2023) and historical information quantum-behaved PSO (HIQPSO) (Chen et al., 2023). The sixth method employs counter-fitting word embeddings and the greedy search to generate adversarial samples, and it is denoted as "TextFooler" (Jin et al., 2020). The last method is known as "CLARE" (Li et al., 2021). It detects the vulnerabilities of a model and deploys masks to the inputs to indicate missing text, and then plugs in an alternative by using a pre-trained MLM. In addition to these static methods, we employ the DynHI method proposed in Section 3.4 as a dynamic baseline. This allows for a direct comparison to further evaluate the variations in the effectiveness of static adversarial samples.

4.2. Evaluation metrics

In addition to our three proposed proprietary evaluation metrics for dynamic attacks presented in the previous section, we also used two traditional metrics for text adversarial attacks in our experiments. They are used to measure the performance of adversarial methods in generating samples at the initial time step. The first one is the attack success rate, denoted by *ASR*. It is used to calculate the percentage of valid samples in the generated sample set $\boldsymbol{X}'$ that satisfy the requirements in Section 2.2, and it is given by,

$$ASR = \left|\boldsymbol{X}^{\text{Valid}}\right| \Big/ \left|\boldsymbol{X}'\right| = \left|\boldsymbol{X}_1^{\text{Valid}}\right| \Big/ \left|\boldsymbol{X}'\right| \tag{6}$$

The second metrics is the average modification rate, denoted by m. It is calculated as the average percentage of added perturbations and is used to measure the efficiency of the attack. Let $\mathcal{E}(X_j', X_j)$ measure the number of different words between the original input sample X_j and the valid adversarial sample X_j' which is transformed from X_j, and let $l(X_j)$ denote the length of X_j. Then, m is calculated by,

$$m = \frac{\sum_{X_j' \in \boldsymbol{X}^{\text{Valid}}} \left[\frac{\mathcal{E}(X_j', X_j)}{l(X_j)}\right]}{\left|\boldsymbol{X}^{\text{Valid}}\right|} \tag{7}$$

4.3. Experimental settings

In our experiments, the victim model BiLSTM used 300-dimensional pre-trained GloVe word embeddings (Pennington et al., 2014), and BERT used the base version. All seven adversarial methods followed parameter settings from their respective literature.

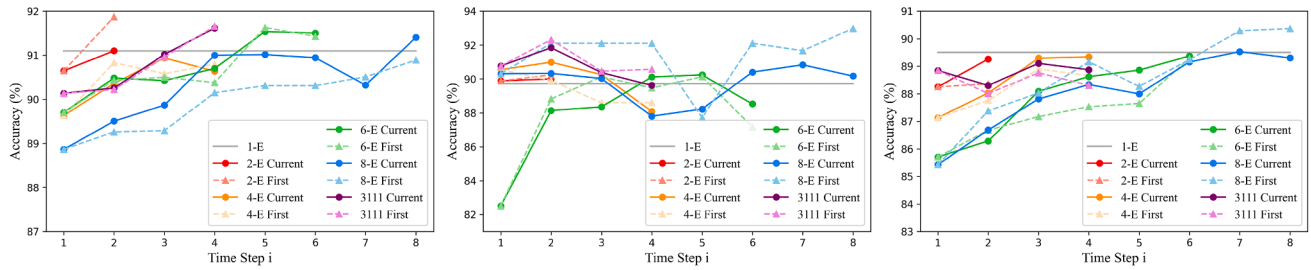
We adopted the same evaluation strategy as in (Alzantot et al., 2018). Input lengths were limited to 10–100. Also, valid samples required a modification rate below 25%; otherwise, the attack was considered a failure. For SNLI, only hypothesis sentences were modified. In the static attack scenario, 1000 correctly classified test samples were randomly selected from all benchmark datasets. However, in the dynamic attack task, due to the splitting of the dataset, sometimes it is hard to find 1000 corrected classified samples to form the original input set. Therefore, the actual number of samples to be attacked in our experiments could only be as large as 1000 as possible.

In our experiments, we designed six splitting schemes with ratio r: five equal splits (1, 2, 4, 6, 8 parts) and one unequal split (3:1:1:1). The 1-equal splitting scheme can be used to evaluate the capability of the adversarial method under the static model. The six splitting schemes are abbreviated as 1,2,4,6,8-E and 3111 for ease of designation.

Alternative splits were not adopted due to some potential issues. For instance, if few samples were used for training during the phase of first training and attacking, it would be hard to produce an effective model and an adequate number of valid adversarial samples. Besides, the performance of the victim continuously updating model would hardly change if few samples were employed for model training at the later phase. Therefore, we exclude these schemes. And the proposed six splitting schemes provide sufficient coverage for evaluating the static text adversarial methods in continuously updating model scenarios.

4.4. Continuously updating model training results and analysis

Fig. 4 shows the training results of the continuously updating BERT model generation. Overall, for both IMDB and SNLI, as the training dataset gradually expanded, the accuracy of the model $\mathcal{M}_i$ on $\boldsymbol{D}_i$ increased progressively, and eventually came to the similar results of the static model (1-E), as the model was ultimately trained on all data. For IMDB, at the initial time step, models trained under splitting schemes with smaller training sets exhibited lower accuracy, but they typically improved quickly and reached the 1-E baseline within a few subsequent updates. SNLI showed a similar pattern to IMDB, yet its model accuracy was more sensitive to the initial training size. Moreover, across different



(a) IMDB (b) SST-2 (c) SNLI

Fig. 4. Training results of the continuously updating BERT model on three datasets under six splitting schemes. "Current" curves indicate the accuracy of model $\mathcal{M}_i$ on $\boldsymbol{D}_i$'s test set, while "First" curves indicate the accuracy of $\mathcal{M}_i$ on $\boldsymbol{D}_1$'s test set. To better show the accuracy of the static model at 1-equal splitting as the benchmark, we extend its corresponding curves.

Table 2
The results of seven adversarial methods of attacking $\mathcal{M}_1$ model under six splitting schemes. %s denotes the attack success rate (%), %m denotes the average modification rate (%).

Dataset	Model	Attack Method	1-E		2-E		4-E		6-E		8-E		3111	
			%s	%m	%s	%m	%s	%m	%s	%m	%s	%m	%s	%m
IMDB	BiLSTM	GA	99.82	6.34	99.62	6.93	99.60	7.01	99.53	6.53	99.02	6.28	99.24	7.12
		PWWS	71.43	6.32	79.10	5.88	76.63	6.22	90.24	6.03	85.12	6.32	75.62	5.78
		DPSO	**99.99**	3.94	**99.99**	3.99	**99.99**	4.71	**99.99**	4.39	**99.99**	4.79	**99.99**	4.51
		DQPSO	**99.99**	3.68	**99.99**	**3.78**	**99.99**	**4.40**	**99.99**	**4.08**	**99.99**	4.52	**99.99**	**4.17**
		HIQPSO	99.82	**3.58**	**99.99**	3.84	**99.99**	4.41	**99.99**	**4.08**	**99.99**	4.47	**99.99**	4.25
		TextFooler	99.80	6.11	99.80	5.86	99.80	6.64	**99.99**	4.74	**99.99**	4.96	99.70	6.64
		CLARE	99.50	4.28	99.70	4.42	99.59	5.08	99.85	4.15	99.34	**4.34**	99.80	4.46
	BERT	GA	90.40	7.59	77.60	7.01	96.00	8.57	82.80	9.59	91.20	9.72	71.60	6.44
		PWWS	62.30	7.57	49.75	4.86	54.30	6.15	58.14	7.07	60.53	6.30	42.14	4.77
		DPSO	99.90	4.55	**99.99**	3.87	**99.99**	3.32	**99.99**	4.38	**99.99**	3.60	**99.99**	3.82
		DQPSO	**99.99**	4.28	**99.99**	3.63	**99.99**	**3.17**	**99.99**	4.19	**99.99**	3.35	**99.99**	3.78
		HIQPSO	99.40	**3.47**	**99.99**	**3.52**	**99.99**	**3.17**	**99.99**	**3.99**	**99.99**	**3.23**	**99.99**	**3.72**
		TextFooler	99.40	10.69	71.90	7.43	98.60	9.44	96.10	11.97	99.14	9.59	71.20	9.08
		CLARE	99.50	4.96	83.90	3.53	99.40	3.98	98.89	4.61	99.57	4.03	81.30	4.14
SST-2	BiLSTM	GA	68.90	12.27	71.05	11.15	64.63	12.84	72.34	11.25	70.59	10.83	67.85	12.51
		PWWS	59.50	15.17	53.15	13.44	59.52	14.27	58.01	14.11	64.80	15.41	61.31	15.95
		DPSO	93.40	10.20	93.67	9.35	92.15	10.11	92.51	9.87	91.85	9.12	93.92	9.94
		DQPSO	94.30	9.90	94.20	9.12	92.49	9.86	94.65	9.83	92.59	**8.93**	94.26	**9.68**
		HIQPSO	94.30	**9.89**	94.20	**9.11**	92.52	**9.84**	93.58	**9.69**	92.59	**8.93**	94.26	9.70
		TextFooler	**97.40**	15.60	**96.58**	14.73	**95.17**	15.44	**97.40**	13.95	**96.65**	15.11	**98.69**	15.02
		CLARE	72.20	11.57	69.73	10.90	72.39	11.00	78.79	11.02	69.83	9.26	72.94	11.18
	BERT	GA	68.80	11.07	69.20	11.43	71.60	10.80	66.50	11.62	75.45	11.36	69.20	11.43
		PWWS	43.50	9.41	42.95	9.77	44.15	9.40	46.60	9.36	54.49	9.37	46.38	8.90
		DPSO	94.40	8.83	95.70	8.39	94.17	8.92	92.24	**8.71**	96.67	8.23	95.03	**8.07**
		DQPSO	94.90	8.68	**96.19**	8.25	**95.79**	8.99	**94.52**	8.96	97.33	8.15	**95.81**	8.08
		HIQPSO	**95.40**	**8.45**	95.84	**7.98**	**95.79**	8.99	94.06	8.92	96.67	**8.06**	**95.81**	**8.07**
		TextFooler	94.20	17.58	93.55	18.51	95.42	17.81	93.88	18.07	**97.99**	15.80	94.64	17.08
		CLARE	71.00	9.96	69.16	10.02	68.96	**8.88**	68.16	9.83	75.38	8.84	68.58	9.18
SNLI	BiLSTM	GA	36.30	13.91	38.70	13.43	23.35	13.09	18.24	13.29	6.25	14.12	37.40	13.29
		PWWS	31.90	16.23	32.10	16.04	31.40	17.66	29.00	17.07	16.41	16.02	30.00	17.37
		DPSO	66.40	11.98	64.80	**11.57**	59.14	12.80	51.98	13.28	17.61	16.65	65.50	11.73
		DQPSO	67.30	**11.85**	63.10	11.66	56.42	12.80	52.28	13.22	15.34	15.84	64.40	11.86
		HIQPSO	64.20	12.18	65.90	11.87	59.92	**12.76**	51.37	**12.74**	21.02	15.61	64.80	**11.72**
		TextFooler	**78.60**	18.51	**81.30**	18.42	**78.80**	18.63	**76.40**	17.88	**47.66**	21.22	**81.80**	18.89
		CLARE	22.20	14.47	21.90	13.61	22.40	14.20	23.20	13.57	14.53	**13.74**	23.10	13.91
	BERT	GA	36.80	12.83	46.10	12.13	45.59	12.27	45.22	11.93	43.63	11.83	40.20	12.46
		PWWS	58.00	**10.73**	64.10	**9.82**	63.18	**10.10**	60.77	**10.43**	62.31	**9.59**	59.50	**10.55**
		DPSO	77.50	11.89	79.30	11.35	**84.56**	11.50	77.25	11.47	77.99	11.67	77.80	11.61
		DQPSO	79.70	12.01	80.90	11.31	83.27	11.32	77.81	11.81	**81.47**	11.73	80.00	11.93
		HIQPSO	79.80	11.84	81.40	11.46	84.19	11.58	77.25	11.97	77.99	11.76	78.50	11.74
		TextFooler	**80.10**	17.51	**82.30**	18.05	81.90	17.72	**82.60**	17.85	79.34	18.15	**81.00**	18.05
		CLARE	22.90	12.46	24.10	13.19	24.40	12.25	25.40	13.28	26.89	12.71	22.40	12.63

splitting schemes, models generally required more training time steps to eventually improve their accuracy to the 1-E baseline.

For SST-2, on the other hand, the training results were more fluctuating. In particular, the accuracy for test set by 6-E was significantly lower than those of the other splitting schemes at the initial time step. This is because the smaller SST-2 dataset made the model more prone to overfitting after splitting. In addition, the shorter average length of sentences in the SST-2 may also exacerbate model overfitting.

Meanwhile, since the static adversarial samples were only generated at $i = 1$, the difference between the accuracy of the current model $\mathcal{M}_i$ on

Table 3
Comparison of attack effectiveness and average generation time (second/valid sample) of different adversarial methods against the BERT model on IMDB.

Method	Time	1-E		2-E		4-E		6-E		8-E		3111	
		%s	%m	%s	%m	%s	%m	%s	%m	%s	%m	%s	%m
GA	772.67s	90.40	7.59	77.60	7.01	96.00	8.57	82.80	9.59	91.20	9.72	71.60	6.44
PWWS	9.95s	62.30	7.57	49.75	4.86	54.30	6.15	58.14	7.07	60.53	6.30	42.14	4.77
DPSO	464.82s	99.90	4.55	**99.99**	3.87	**99.99**	3.32	**99.99**	4.38	**99.99**	3.60	**99.99**	3.82
DQPSO	471.96s	**99.99**	4.28	**99.99**	3.63	**99.99**	**3.17**	**99.99**	4.19	**99.99**	3.35	**99.99**	3.78
HIQPSO	478.07s	99.40	**3.47**	**99.99**	**3.52**	**99.99**	**3.17**	**99.99**	**3.99**	**99.99**	**3.23**	**99.99**	**3.72**
TextFooler	**6.65**s	99.40	10.69	71.90	7.43	98.60	9.44	96.10	11.97	99.14	9.59	71.20	9.08
CLARE	23.73s	99.50	4.96	83.90	3.53	99.40	3.98	98.89	4.61	99.57	4.03	81.30	4.14

$\boldsymbol{D}_1$ and on $\boldsymbol{D}_i$ must be explored. On both IMDB and SNLI, the accuracies on $\boldsymbol{D}_i$ and on $\boldsymbol{D}_1$ maintained a similar upward trend. The reason of this may be that the chosen datasets were evenly distributed and had been disordered in advance. In particular, under 2-E, the difference between the accuracies on $\boldsymbol{D}_i$ and on $\boldsymbol{D}_1$ was larger for these two datasets at $i = 2$. And compared to 3111, for each dataset, the difference between the accuracies on $\boldsymbol{D}_i$ and on $\boldsymbol{D}_1$ by 2-E was much smaller when $\boldsymbol{D}_1$ was of the same size in two splitting schemes (both half of the original dataset). Therefore, we believe that excessive data at later time steps, when the initial data is already abundant, may lead to unbalanced model learning. This means that it would cause the model prediction to be easily biased to a certain data distribution (Wang et al., 2021). In addition, in the cases of 6-E and 8-E, the difference between the accuracies on $\boldsymbol{D}_i$ and on $\boldsymbol{D}_1$ for the two datasets also appeared to be large. The models in these two cases had more training steps, so that the distribution of training data changed more. Consequently, although the accuracy gradually increased, the prediction distribution was actually prone to change.

For SST-2, the difference between the accuracies on $\boldsymbol{D}_i$ and on $\boldsymbol{D}_1$ was larger in 4-E, 6-E, and 8-E, which implied the overfitting of the model since the dataset is smaller.

4.5. Static attack results and analysis

The results of the static adversarial methods in attacking $\mathcal{M}_1$ (i.e., static attacks) to generate static adversarial samples are shown in Table 2. In general, experiments showed that model accuracy varied across splits, but robustness to adversarial attacks remained consistent. In most cases, the adversarial methods responded slowly to changes in model accuracy, indicating that such variations in accuracy had little effect on attack effectiveness. As for the three methods based on sememes and PSO, they were stable and achieved strong attack performance.

For the BiLSTM model on SNLI, attack performance slightly declined with less data at the initial time step. This can be partly attributed to the gradual decline of model accuracy. Meanwhile, the short average sentence length in SNLI could lead to over-modification by the adversarial methods, making attacks increasingly difficult as the amount of initial data decreased. In particular, all methods performed the worst in 8-E. In addition to extremely low model accuracy (only 53.6%) and short sentence length, the small size of the test set at $i = 1$ was also a contributing factor, which limited the generation of sufficient adversarial samples. These three factors together caused a significant decrease in the effectiveness of the adversarial methods.

4.6. Static attack time consumption analysis

Table 3 presents the average time taken by different adversarial methods to generate one valid adversarial sample against the BERT model on IMDB, along with their attack effectiveness. It can be observed that PWWS consumes very little time but attains the lowest attack effectiveness, limiting its practical utility. DPSO, DQPSO, and HIQPSO achieve the best attack effectiveness and the most stable attack results, but their excessive time cost makes them unsuitable for real-time attacks. TextFooler is the fastest method, yet it exhibits the highest perturbation rate and relatively unstable attack success rates. CLARE provides a more balanced trade-off between time and attack performance. However, its effectiveness declines significantly on the SST and SNLI datasets, with especially poor performance on SNLI (Table 2), indicating that it cannot serve as a general solution balancing efficiency and effectiveness.

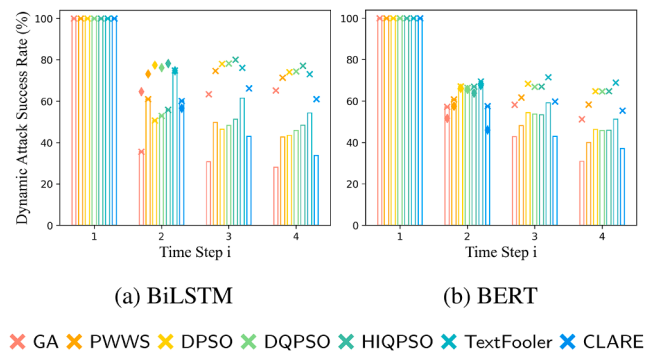


Fig. 5. Results for single and continuous dynamic attack success rates on SNLI under 3111 split. The x markers and bars represent SDR_i and CDR_i, respectively. Note that $SDR_i = CDR_i$ at $i = 1$ *or* 2. And for better comparison, diamond-shaped markers are used to represent the single dynamic attack success rates at the second time step under 2-E split.

In summary, the existing static adversarial methods fail to achieve both strong attack effectiveness and fast generation speed simultaneously. Since generating a large number of valid adversarial samples after each model update is infeasible due to the high time cost, we instead perform adversarial attacks only on the initial model $\mathcal{M}_1$ to obtain a static set of valid adversarial samples $\boldsymbol{X}^{\text{Valid}}$. These samples are then used to attack subsequent models in order to evaluate the effectiveness of static adversarial methods against continuously updating models.

4.7. Dynamic attack results and analysis

4.7.1. Single dynamic attack

For single dynamic attack, the attack results for the continuously updating models with static valid samples with each splitting scheme are shown in Table 4. It can be observed that SDR_i sharply declined since $i = 2$, followed by fluctuations within a certain range or slight declines. To explore the specific reasons for the changes in SDR_i, we made an analysis of the results by the two splits: 2-E and 3111. Partial results are shown in Fig. 5.

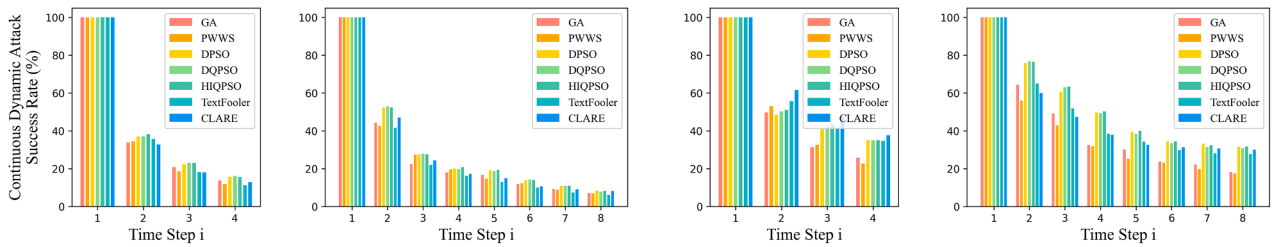
By comparing the results of 3111 and 2-E after the initial time step, it was found that the SDR_i of 3111 fluctuated around that of 2-E. In the attack on the BERT model for SNLI, as in Fig. 5, the SDR_i of 3111 and 2-E was similar after the initial time step. However, in the attacks on the BiLSTM model for SNLI, the SDR_i of 3111 at $i = 2$ was significantly worse than that of 2-E, but later reached a level similar to 2-E. Therefore, we can conclude that when the continuously updating model was

Table 4
Single dynamic attack success rate (SDR, %) under all splitting schemes.

Dataset	Model	Attack Method	2-E SDR at i		4-E SDR at i				6-E SDR at i						8-E SDR at i								3111 SDR at i			
			1	2	1	2	3	4	1	2	3	4	5	6	1	2	3	4	5	6	7	8	1	2	3	4
IMDB	BiLSTM	GA	100.00	47.14	100.00	50.40	45.75	42.71	100.00	41.22	39.11	40.75	37.94	43.33	100.00	50.66	43.38	40.40	39.07	38.41	41.06	35.43	100.00	43.59	49.90	56.79
		PWWS	100.00	39.32	100.00	33.69	33.42	33.29	100.00	32.60	27.87	26.86	28.38	31.93	100.00	44.73	41.13	37.53	33.68	37.79	34.96	29.05	100.00	42.67	43.73	46.63
		DPSO	100.00	44.70	100.00	42.03	38.13	38.91	100.00	39.25	34.35	33.88	32.48	40.89	100.00	45.90	34.43	33.44	34.10	38.36	38.69	31.48	100.00	46.70	47.70	53.80
		DQPSO	100.00	45.60	100.00	44.84	38.28	39.22	100.00	37.38	34.58	33.64	31.54	41.59	100.00	49.51	36.72	36.07	34.10	39.67	42.62	35.08	100.00	46.55	48.05	54.45
		HIQPSO	100.00	42.67	100.00	43.43	36.77	40.20	100.00	37.85	36.45	35.75	33.41	42.52	100.00	47.87	36.07	36.72	35.08	40.66	40.98	34.43	100.00	41.44	48.29	52.28
		TextFooler	100.00	38.08	100.00	39.57	37.12	33.44	100.00	36.74	30.18	25.91	29.88	31.10	100.00	44.86	37.20	36.11	36.32	33.70	32.82	31.73	100.00	43.33	46.34	49.55
		CLARE	100.00	36.31	100.00	34.94	34.32	33.50	100.00	27.02	25.50	28.40	30.99	29.47	100.00	42.73	37.67	38.77	31.06	37.00	36.56	31.72	100.00	38.48	38.58	41.98
		DynHI	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.63	100.00	99.64	100.00	100.00	100.00	99.62	100.00	100.00	99.78	100.00
	BERT	GA	100.00	30.41	100.00	33.75	33.33	35.00	100.00	44.93	46.38	47.34	34.78	41.06	100.00	44.30	35.52	41.23	42.98	32.46	39.91	34.21	100.00	33.52	30.73	18.99
		PWWS	100.00	20.95	100.00	34.47	29.19	35.71	100.00	35.64	36.00	37.09	31.27	34.91	100.00	42.65	38.24	32.35	33.33	36.76	31.37	31.37	100.00	32.79	39.75	32.79
		DPSO	100.00	38.23	100.00	37.02	34.43	36.59	100.00	48.70	53.04	52.17	52.61	45.65	100.00	52.38	42.18	40.48	47.62	42.52	40.82	31.63	100.00	46.23	42.84	42.95
		DQPSO	100.00	39.53	100.00	37.02	35.58	37.02	100.00	50.87	56.52	53.70	56.74	46.74	100.00	53.06	40.48	42.19	45.24	43.20	40.14	32.99	100.00	46.62	43.79	43.90
		HIQPSO	100.00	38.27	100.00	38.23	34.87	36.99	100.00	52.07	55.12	53.16	54.68	46.41	100.00	52.38	41.16	43.54	47.28	42.52	40.82	33.67	100.00	50.22	45.41	45.41
		TextFooler	100.00	34.49	100.00	35.70	29.72	33.57	100.00	48.99	51.45	46.09	46.81	41.45	100.00	41.52	36.96	33.26	37.83	37.39	39.13	34.57	100.00	48.31	35.39	33.29
		CLARE	100.00	36.95	100.00	32.80	30.18	35.11	100.00	46.76	48.45	44.37	39.44	37.46	100.00	46.97	36.36	33.98	38.31	33.55	30.52	32.03	100.00	46.74	34.93	37.88
		DynHI	100.00	99.76	100.00	99.79	99.80	100.00	100.00	99.55	99.77	99.77	99.77	99.78	100.00	100.00	100.00	100.00	99.64	99.65	99.28	99.64	100.00	97.99	97.98	97.99
SST-2	BiLSTM	GA	100.00	47.65	100.00	59.47	55.79	55.26	100.00	54.41	43.38	33.82	37.50	38.97	100.00	47.92	44.79	50.00	45.83	38.54	44.79	37.50	100.00	51.36	57.07	54.34
		PWWS	100.00	52.58	100.00	56.31	46.40	46.40	100.00	51.49	36.57	33.58	36.57	38.81	100.00	50.00	48.28	50.86	43.97	44.83	39.66	43.10	100.00	51.60	63.97	60.34
		DPSO	100.00	53.28	100.00	65.19	55.56	52.59	100.00	66.47	48.56	43.35	44.51	49.71	100.00	52.42	54.03	54.84	44.36	46.77	42.74	43.55	100.00	58.27	62.23	59.71
		DQPSO	100.00	55.41	100.00	65.68	54.98	53.14	100.00	67.80	51.41	46.33	45.76	49.15	100.00	53.60	53.60	53.60	44.00	45.60	41.60	39.20	100.00	58.42	63.98	59.14
		HIQPSO	100.00	54.85	100.00	65.07	54.41	52.57	100.00	68.57	52.00	45.71	45.71	49.14	100.00	52.80	54.40	56.80	44.80	46.40	43.20	40.80	100.00	58.42	63.62	58.78
		TextFooler	100.00	55.60	100.00	67.89	58.59	58.59	100.00	64.44	48.89	44.00	47.11	47.56	100.00	58.96	61.27	55.49	54.91	50.87	53.18	52.02	100.00	63.18	68.87	66.09
		CLARE	100.00	54.42	100.00	59.63	56.67	55.93	100.00	62.64	49.45	42.31	46.70	46.15	100.00	48.00	46.40	50.40	44.80	50.40	44.00	44.00	100.00	59.32	64.70	61.65
		DynHI	100.00	94.80	100.00	94.76	95.12	94.63	100.00	92.86	92.59	90.68	93.33	91.88	100.00	92.79	95.41	94.64	94.06	93.33	92.59	95.45	100.00	93.13	95.13	93.73
	BERT	GA	100.00	54.91	100.00	49.72	48.04	51.95	100.00	59.54	50.38	58.78	54.20	45.80	100.00	64.28	64.28	57.94	60.32	55.56	54.00	52.38	100.00	57.80	56.63	52.02
		PWWS	100.00	51.87	100.00	53.03	42.42	41.67	100.00	45.83	42.71	40.62	37.50	35.42	100.00	56.04	63.74	59.34	54.95	58.24	58.24	49.45	100.00	48.96	52.08	51.39
		DPSO	100.00	61.42	100.00	48.45	60.48	53.95	100.00	64.85	65.35	60.89	67.33	52.49	100.00	75.86	68.28	62.46	50.34	57.24	64.14	61.38	100.00	62.75	53.27	53.43
		DQPSO	100.00	62.48	100.00	50.34	59.46	53.04	100.00	63.29	64.25	57.97	66.18	52.17	100.00	76.71	69.86	61.64	51.37	58.22	64.38	61.64	100.00	63.05	53.32	53.81
		HIQPSO	100.00	60.76	100.00	51.01	59.12	52.03	100.00	63.11	64.08	57.77	66.02	51.94	100.00	76.55	70.34	62.76	52.41	58.62	64.83	62.07	100.00	62.88	53.32	54.13
		TextFooler	100.00	67.59	100.00	55.73	57.33	53.87	100.00	67.83	63.04	64.78	62.17	54.35	100.00	65.13	63.59	58.46	58.46	60.51	63.08	60.51	100.00	58.57	61.94	60.46
		CLARE	100.00	65.67	100.00	61.62	59.04	56.09	100.00	61.68	58.68	61.68	66.47	58.68	100.00	60.00	62.67	64.00	55.33	62.00	61.33	63.33	100.00	69.27	64.80	62.38
		DynHI	100.00	96.18	100.00	98.08	97.67	97.66	100.00	93.02	94.77	94.08	91.62	94.05	100.00	96.05	94.00	96.45	95.17	95.89	96.64	96.03	100.00	95.94	94.10	96.26
SNLI	BiLSTM	GA	100.00	64.60	100.00	61.67	59.17	71.67	100.00	40.00	40.00	38.33	36.67	40.00	100.00	36.36	45.45	45.45	54.55	45.45	36.36	36.36	100.00	35.56	63.37	65.24
		PWWS	100.00	73.21	100.00	62.10	57.96	54.46	100.00	56.55	53.79	52.41	47.93	47.24	100.00	34.29	38.10	35.24	34.29	27.62	26.67	29.52	100.00	61.00	74.67	71.33
		DPSO	100.00	77.47	100.00	73.68	76.32	83.55	100.00	70.76	72.51	67.25	70.18	63.16	100.00	74.19	45.16	45.16	41.94	51.61	48.39	48.39	100.00	50.69	78.02	74.05
		DQPSO	100.00	76.23	100.00	74.48	74.14	81.72	100.00	73.84	73.26	68.02	68.02	62.79	100.00	74.07	62.96	66.67	55.56	51.85	55.56	44.44	100.00	52.95	78.26	74.38
		HIQPSO	100.00	78.30	100.00	76.95	76.30	83.44	100.00	72.78	73.37	65.68	68.64	61.54	100.00	78.38	56.76	62.16	59.46	54.05	45.95	37.84	100.00	55.86	80.09	77.16
		TextFooler	100.00	74.78	100.00	63.32	61.68	56.85	100.00	58.90	55.37	54.84	46.60	47.91	100.00	41.64	40.66	39.02	40.66	35.08	36.39	36.07	100.00	74.82	76.16	73.11
		CLARE	100.00	56.62	100.00	52.68	52.68	42.41	100.00	44.40	42.24	39.66	40.95	34.48	100.00	25.26	12.63	13.68	20.00	17.89	15.79	23.16	100.00	60.17	66.23	61.04
		DynHI	100.00	96.75	100.00	96.72	95.64	95.92	100.00	95.60	96.18	94.94	93.46	96.91	100.00	100.00	91.30	86.36	92.00	90.48	90.48	90.91	100.00	90.34	97.38	96.91
	BERT	GA	100.00	51.63	100.00	46.37	45.56	42.34	100.00	47.83	49.69	42.24	39.75	37.89	100.00	37.17	51.33	45.13	38.94	30.97	33.63	33.63	100.00	57.21	58.21	51.24
		PWWS	100.00	57.57	100.00	59.67	56.91	51.38	100.00	57.73	59.55	46.36	53.18	52.73	100.00	40.12	55.09	48.50	52.10	53.89	49.10	49.10	100.00	60.84	61.68	58.32
		DPSO	100.00	66.20	100.00	60.22	55.22	54.78	100.00	68.73	64.73	54.55	54.55	53.82	100.00	50.50	65.84	60.40	53.96	57.92	55.94	48.02	100.00	67.10	68.38	64.78
		DQPSO	100.00	65.64	100.00	61.81	55.19	55.41	100.00	67.15	67.15	55.60	53.43	53.07	100.00	46.92	62.09	55.92	51.66	55.45	52.61	46.92	100.00	66.38	66.88	64.63
		HIQPSO	100.00	63.88	100.00	60.92	55.24	55.46	100.00	64.00	64.36	52.36	51.64	51.27	100.00	46.53	63.86	56.44	50.00	55.94	50.99	44.55	100.00	67.01	67.01	64.71
		TextFooler	100.00	67.56	100.00	64.96	63.37	62.88	100.00	64.65	65.50	62.83	65.86	62.47	100.00	56.15	58.25	58.12	58.51	54.45	54.84	54.97	100.00	69.51	71.48	68.89
		CLARE	100.00	46.06	100.00	49.18	45.49	42.62	100.00	42.52	48.82	43.31	41.73	40.16	100.00	37.69	40.38	36.92	41.92	40.38	40.77	38.46	100.00	57.59	59.82	55.36
		DynHI	100.00	94.41	100.00	93.02	88.44	92.04	100.00	93.36	89.60	92.31	85.43	91.97	100.00	85.20	84.82	86.22	87.30	90.58	90.00	88.27	100.00	95.30	93.53	94.49

Table 5
Continuous dynamic attack success rate (CDR, %) under all splitting schemes.

Dataset	Model	Attack Method	2-E CDR at i		4-E CDR at i				6-E CDR at i						8-E CDR at i								3111 CDR at i			
			1	2	1	2	3	4	1	2	3	4	5	6	1	2	3	4	5	6	7	8	1	2	3	4
IMDB	BiLSTM	GA	100.00	47.14	100.00	50.40	32.39	24.09	100.00	41.22	24.36	19.44	15.22	11.94	100.00	50.66	32.12	23.18	15.56	11.59	7.95	6.62	100.00	43.59	26.20	22.75
		PWWS	100.00	39.32	100.00	33.69	20.37	15.31	100.00	32.60	17.06	12.67	9.80	8.61	100.00	44.73	26.22	19.02	11.31	8.48	6.17	4.63	100.00	42.67	24.44	18.23
		DPSO	100.00	44.70	100.00	42.03	25.00	20.16	100.00	39.25	23.13	18.22	14.25	11.21	100.00	45.90	23.93	17.05	11.48	9.84	7.87	7.21	100.00	46.70	26.90	21.50
		DQPSO	100.00	45.60	100.00	44.84	26.56	21.72	100.00	37.38	22.90	17.06	13.08	10.28	100.00	49.51	26.23	19.02	12.13	10.82	9.51	7.87	100.00	46.55	27.33	23.02
		HIQPSO	100.00	42.67	100.00	43.43	26.06	20.61	100.00	37.85	23.83	17.99	12.85	10.05	100.00	47.87	25.57	18.03	12.79	10.49	8.52	6.56	100.00	41.44	23.76	19.96
		TextFooler	100.00	38.08	100.00	39.57	24.95	18.10	100.00	36.74	18.90	12.35	8.99	6.40	100.00	44.86	26.48	15.97	10.50	7.44	5.69	4.16	100.00	43.33	24.97	18.86
		CLARE	100.00	36.31	100.00	34.94	21.41	16.09	100.00	27.02	14.05	11.15	9.16	6.41	100.00	42.73	22.47	17.40	10.35	8.59	5.95	5.51	100.00	38.48	20.74	15.43
		DynHI	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	99.71	99.41	100.00	99.63	99.25	98.84	98.41	97.94	97.45	96.95	100.00	100.00	99.77	99.52
	BERT	GA	100.00	30.41	100.00	33.75	20.83	13.75	100.00	44.93	32.37	28.99	21.74	17.39	100.00	44.30	22.37	17.98	16.67	11.84	9.21	7.02	100.00	33.52	19.55	8.38
		PWWS	100.00	20.95	100.00	34.47	18.63	11.80	100.00	35.64	22.18	13.82	12.00	9.45	100.00	42.65	27.45	19.61	14.71	12.25	8.82	6.86	100.00	32.79	22.13	17.21
		DPSO	100.00	38.23	100.00	37.02	22.38	15.78	100.00	48.70	38.91	35.43	30.87	25.00	100.00	52.38	27.55	20.07	19.05	13.95	10.88	8.50	100.00	46.23	28.96	22.51
		DQPSO	100.00	39.53	100.00	37.02	22.96	16.07	100.00	50.87	40.87	37.39	33.26	27.17	100.00	53.06	27.89	19.73	18.71	14.29	10.88	7.82	100.00	46.62	29.74	23.09
		HIQPSO	100.00	38.27	100.00	38.23	23.01	15.58	100.00	52.07	42.05	37.91	32.68	26.80	100.00	52.38	27.55	20.75	19.39	13.95	10.88	8.16	100.00	50.22	31.88	24.89
		TextFooler	100.00	34.49	100.00	35.70	18.15	11.26	100.00	48.99	40.00	34.64	30.00	23.33	100.00	41.52	21.96	16.30	13.04	10.00	7.39	6.09	100.00	48.31	23.46	16.01
		CLARE	100.00	36.95	100.00	32.80	18.01	12.88	100.00	46.76	33.94	27.75	20.00	16.06	100.00	46.97	24.46	17.10	14.94	10.61	9.09	8.23	100.00	46.74	23.74	19.07
		DynHI	100.00	99.76	100.00	99.79	99.58	99.36	100.00	99.55	99.32	99.09	98.40	97.46	100.00	100.00	100.00	99.62	99.24	98.86	98.08	96.92	100.00	97.99	95.95	93.94
SST-2	BiLSTM	GA	100.00	47.65	100.00	59.47	45.79	40.53	100.00	54.41	33.82	21.32	16.91	16.18	100.00	47.92	34.38	20.83	16.67	12.50	12.50	8.33	100.00	51.36	37.97	30.27
		PWWS	100.00	52.58	100.00	56.31	38.74	31.53	100.00	51.49	28.36	21.64	17.16	14.93	100.00	50.00	35.34	21.55	16.38	12.93	10.34	10.34	100.00	51.60	40.51	33.69
		DPSO	100.00	53.28	100.00	65.19	49.63	40.37	100.00	66.47	42.77	29.48	24.28	21.39	100.00	52.42	41.13	25.81	19.35	16.94	16.94	13.71	100.00	58.27	44.06	36.69
		DQPSO	100.00	55.41	100.00	65.68	49.45	41.33	100.00	67.80	45.76	31.07	25.42	22.60	100.00	53.60	40.80	28.80	22.40	20.00	17.60	13.60	100.00	58.42	44.60	36.20
		HIQPSO	100.00	54.85	100.00	65.07	48.90	40.44	100.00	68.57	46.86	32.00	26.29	23.43	100.00	52.80	40.80	29.60	23.20	20.80	18.40	14.40	100.00	58.42	44.80	36.20
		TextFooler	100.00	55.60	100.00	67.89	50.70	44.23	100.00	64.44	42.22	29.33	24.89	21.33	100.00	58.96	45.09	31.21	26.59	23.70	21.97	19.65	100.00	63.18	50.46	44.24
		CLARE	100.00	54.42	100.00	59.63	47.41	40.37	100.00	62.64	42.31	29.12	24.18	21.43	100.00	48.00	34.40	22.40	18.40	16.80	13.60	10.40	100.00	59.32	46.95	39.07
		DynHI	100.00	94.80	100.00	94.76	88.34	81.94	100.00	92.86	84.92	74.89	65.13	56.44	100.00	92.79	84.61	74.91	65.77	57.44	48.49	40.74	100.00	93.13	86.15	78.06
	BERT	GA	100.00	54.91	100.00	49.72	31.28	25.70	100.00	59.54	42.75	36.64	32.06	25.95	100.00	64.28	49.21	32.54	30.16	23.81	22.22	18.25	100.00	57.80	46.82	32.37
		PWWS	100.00	51.87	100.00	53.03	32.58	22.73	100.00	45.83	33.33	21.88	18.75	13.54	100.00	56.04	42.86	31.87	25.27	23.08	19.78	17.58	100.00	48.96	34.72	27.43
		DPSO	100.00	61.42	100.00	48.45	40.89	35.05	100.00	64.85	50.50	43.56	40.59	32.18	100.00	75.86	60.69	49.66	39.31	34.48	33.10	31.41	100.00	62.75	46.41	36.60
		DQPSO	100.00	62.48	100.00	50.34	42.57	35.14	100.00	63.29	48.79	40.58	38.16	31.40	100.00	76.71	63.01	49.32	38.36	33.56	31.51	30.82	100.00	63.05	47.00	37.43
		HIQPSO	100.00	60.76	100.00	51.01	43.24	35.14	100.00	63.11	48.54	40.29	37.86	31.07	100.00	76.55	63.45	50.34	40.00	34.48	32.41	31.72	100.00	62.88	46.84	37.12
		TextFooler	100.00	67.59	100.00	55.73	41.87	34.67	100.00	67.83	49.13	42.61	38.26	32.17	100.00	65.13	51.79	38.46	34.36	29.74	28.21	27.69	100.00	58.57	48.72	39.81
		CLARE	100.00	65.67	100.00	61.62	47.23	37.64	100.00	61.68	44.91	36.53	33.53	27.54	100.00	60.00	47.33	38.00	32.67	31.33	30.67	30.00	100.00	69.27	55.31	45.62
		DynHI	100.00	96.18	100.00	98.08	95.67	92.45	100.00	93.02	84.78	76.87	68.81	62.32	100.00	96.05	89.47	82.02	74.80	68.00	61.82	55.69	100.00	95.94	91.07	85.76
SNLI	BiLSTM	GA	100.00	64.60	100.00	61.67	52.50	50.83	100.00	40.00	30.00	21.67	16.67	11.67	100.00	36.36	27.27	18.18	18.18	18.18	18.18	18.18	100.00	35.56	30.75	28.07
		PWWS	100.00	73.21	100.00	62.10	52.55	44.59	100.00	56.55	48.97	40.69	34.83	31.72	100.00	34.29	27.62	24.76	23.81	19.05	16.19	15.24	100.00	61.00	49.67	42.67
		DPSO	100.00	77.47	100.00	73.68	70.07	67.43	100.00	70.76	66.08	56.73	54.97	47.95	100.00	74.19	45.16	41.93	38.71	38.71	32.26	29.03	100.00	50.69	46.41	43.51
		DQPSO	100.00	76.23	100.00	74.48	68.62	65.86	100.00	73.84	66.86	55.81	52.33	47.67	100.00	74.07	59.26	49.15	40.74	40.74	37.03	25.93	100.00	52.95	48.29	45.81
		HIQPSO	100.00	78.30	100.00	76.95	70.78	68.51	100.00	72.78	67.46	56.80	55.03	48.52	100.00	78.38	56.76	48.65	45.95	40.54	35.14	27.03	100.00	55.86	51.39	48.46
		TextFooler	100.00	74.78	100.00	63.32	55.33	46.95	100.00	58.90	50.00	43.19	36.39	34.29	100.00	41.64	32.13	30.82	29.51	22.62	20.00	19.02	100.00	74.82	61.49	54.28
		CLARE	100.00	56.62	100.00	52.68	46.43	33.48	100.00	44.40	36.21	28.88	24.14	22.84	100.00	25.26	10.53	10.53	10.53	9.47	8.42	8.42	100.00	60.17	42.86	33.77
		DynHI	100.00	96.75	100.00	96.72	94.47	93.21	100.00	95.60	94.29	92.92	92.20	92.20	100.00	100.00	90.00	83.57	83.57	83.57	83.57	83.57	100.00	90.34	89.81	88.54
	BERT	GA	100.00	51.63	100.00	46.37	33.87	26.21	100.00	47.83	34.78	27.95	22.98	21.12	100.00	37.17	32.74	30.09	26.55	20.35	15.93	12.39	100.00	57.21	42.79	30.85
		PWWS	100.00	57.57	100.00	59.67	45.03	32.60	100.00	57.73	45.91	32.27	29.55	26.82	100.00	40.12	33.53	28.74	24.55	21.56	20.36	19.16	100.00	60.84	48.07	40.00
		DPSO	100.00	66.20	100.00	60.22	46.96	37.83	100.00	68.73	55.64	42.18	34.91	31.27	100.00	50.50	47.52	41.09	36.14	33.17	30.69	25.25	100.00	67.10	54.37	46.27
		DQPSO	100.00	65.64	100.00	61.81	47.02	37.97	100.00	67.15	55.23	41.88	35.38	31.41	100.00	46.92	43.60	39.34	32.70	29.86	27.49	23.22	100.00	66.38	53.75	45.88
		HIQPSO	100.00	63.88	100.00	60.92	45.85	37.99	100.00	64.00	50.55	37.09	30.55	27.64	100.00	46.53	42.57	38.12	30.69	27.72	24.75	20.30	100.00	67.01	53.38	45.99
		TextFooler	100.00	67.56	100.00	64.96	51.77	45.18	100.00	64.65	52.42	45.28	42.74	38.62	100.00	56.15	45.68	40.31	34.42	31.02	28.40	26.18	100.00	69.51	59.14	51.23
		CLARE	100.00	46.06	100.00	49.18	33.61	25.82	100.00	42.52	30.31	23.23	20.47	18.50	100.00	37.69	27.69	20.77	16.92	16.15	15.00	12.31	100.00	57.59	42.86	37.05
		DynHI	100.00	94.41	100.00	93.02	85.98	84.45	100.00	93.36	85.47	83.21	76.70	74.80	100.00	85.20	81.96	79.21	77.45	75.62	75.01	71.91	100.00	95.30	90.25	88.28



(a) IMDB under 4-E (b) IMDB under 8-E (c) SST-2 under 4-E (d) SST-2 under 8-E

Fig. 6. Results for continuous dynamic attack success rates on BERT of IMDB and SST-2 under 4-E and 8-E splits.

trained with additional data at subsequent time steps, its predictive distribution continued to shift around the distribution from the initial time step. Since this shift was influenced by the distribution of the subsequent data and exhibits randomness, it led to fluctuations of SDR_i of 3111 in the subsequent time steps. Despite the different amounts of data obtained by these two splitting schemes at subsequent time steps, SDR_i of 3111 still remained close to that of 2-E, which supported the aforementioned conclusion. Additionally, other SDR_i of equal splitting schemes also showed a downward trend. One reason for this is that the accuracy of model improved. The other one is that the amount of data the model received each time was equal to that at the initial time step.

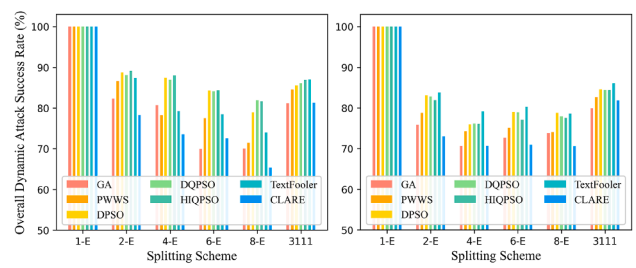
Furthermore, the results of the lightweight dynamic baseline DynHI in Table 4 provided a critical contrast to the static methods. While static methods suffered significant SDR_i degradation, DynHI maintained SDR_i values around 90% or even higher across all splitting schemes, reaching nearly 100% on IMDB. This gap indicated that the decline in static methods was not because continuously updating models had become more resistant to adversarial perturbations. Instead, it reflected an inherent limitation of static adversarial samples in that they failed to adapt to the subtle shifts in the decision boundary of the model. The high success of DynHI's "warm-start" re-optimization confirmed that a substantial adversarial space remained available after model updates, which static methods failed to exploit due to their lack of dynamic adaptability.

4.7.2. Continuous dynamic attack

Then, to further explore the dynamic attack ability of static samples, we conducted continuous dynamic attacks. In this attack experiments, only the samples whose attack was valid on the previous model $\mathcal{M}_{i-1}$ were used to attack the current model $\mathcal{M}_i$. The continuous dynamic attack results are shown in Table 5 and Fig. 6. It can be observed that as i increased, CDR_i decreased in a curve like an inverse proportional function and became stable at the end, indicating that these adversarial samples were of higher quality and more effective in attacking the model compared to other samples.

To investigate the reasons for the changes in CDR_i, we further made a comparative analysis between 3111 and 2-E. They obtained the same amount of data at the initial time step, and both obtained complete data at the final time step. Nevertheless, the CDR_i of 3111 kept decreasing and was significantly lower than that of 2-E at the final time step. This again illustrates that the continuously updating model shifts its prediction distribution after each iteration.

We compared the SDR_i and CDR_i of static methods in most of the cases and show the comparison for some cases of BERT and BiLSTM in Fig. 5. It is found that as i increased, CDR_i consistently decreased regardless of changes of SDR_i. The SDR_i is for directly measuring the gap between $\mathcal{M}_i$ and $\mathcal{M}_1$, whereas the CDR_i for capturing the cumulative effect of changes in the continuously updating model. The difference between SDR_i and CDR_i is actually the difference between the samples that can only superficially confuse the current model and those that can attack the continuously updating model with higher quality.



(a) BiLSTM (b) BERT

Fig. 7. Overall dynamic attack success rate results on SNLI.

In addition, from the facts that SDR_i remained stable and CDR_i gradually decreased over subsequent time steps (as shown in Fig. 5), we believe that the enhancement of the model robustness was actually caused by the change in its accuracy and the shift of its predictive distribution. This is because that if the overall robustness of the model was enhanced, SDR_i should also show a decreasing trend, which was not observed in the experiments.

In contrast, unlike the substantial decline of static methods, DynHI exhibited much better persistence, especially on IMDB where CDR_i remained above 93.9%. These results validated that the observed performance decay stemmed from the inherent limitations of static methods rather than an overall enhancement in model robustness. Meanwhile, because DynHI only employed lightweight re-optimization, it remained constrained by the intrinsic attack success rate of the underlying HIQPSO. Thus, a gradual decline in CDR_i of DynHI was expected as i increased.

Finally, we calculated the overall dynamic attack success rate *Overall*, according to Eq. (3)–(5), and the results were shown in Table 6 and Fig. 7. It can be seen that the proposed evaluation framework were able to measure the effectiveness of static text adversarial methods in the scenarios of continuously updating model in a better way. While the differences in *Overall* among various static methods remained minor, the dynamic baseline DynHI consistently achieved values above 80%, significantly outperforming the static methods. Compared with the large gaps in *ASR* observed in Table 2, the narrowed differences and generally lower scores in *Overall* among static methods highlighted their significant deficiency in the face of continuously updating models. This phenomenon reinforced the importance of this work in investigating the effectiveness of the static text adversarial methods in the scenarios of continuously updating model. The fact that continuous training of the model improved accuracy and altered prediction distributions served as the primary reason for the failure of many adversarial samples, a conclusion further validated by DynHI's high *Overall* score in overcoming the static samples' inability to adapt to these updates.

Table 6
Overall dynamic attack success rates (%) under all splitting schemes.

Model	Attack Method	IMDB						SST-2						SNLI					
		1-E	2-E	4-E	6-E	8-E	3111	1-E	2-E	4-E	6-E	8-E	3111	1-E	2-E	4-E	6-E	8-E	3111
BiLSTM	GA	100.00	73.57	70.24	69.37	71.81	78.34	100.00	73.83	76.77	70.38	71.18	80.19	100.00	82.30	80.67	69.99	70.01	81.20
	PWWS	100.00	69.66	65.57	66.00	70.02	76.47	100.00	76.29	73.11	70.21	71.58	81.69	100.00	86.61	78.23	77.53	71.48	84.54
	DPSO	100.00	72.35	68.02	68.55	69.26	77.59	100.00	76.64	77.01	74.05	73.28	82.00	100.00	88.74	87.49	84.29	78.89	85.60
	DQPSO	100.00	72.80	68.71	68.09	70.16	78.06	100.00	77.71	77.33	74.75	74.25	81.92	100.00	88.12	86.94	84.15	81.92	86.11
	HIQPSO	100.00	71.34	68.23	68.54	69.77	77.46	100.00	77.43	76.97	75.09	74.58	81.89	100.00	89.15	88.02	84.34	81.65	86.87
	TextFooler	100.00	69.04	66.99	66.08	68.87	76.84	100.00	77.80	78.70	73.96	75.66	84.34	100.00	87.39	79.26	78.46	73.98	87.01
	CLARE	100.00	68.16	65.94	65.26	69.37	75.51	100.00	77.21	76.80	73.78	71.88	82.78	100.00	78.31	73.61	72.56	65.40	81.32
	DynHI	100.00	**100.00**	**100.00**	**99.85**	**99.21**	**99.87**	100.00	**97.40**	**94.42**	**86.92**	**82.10**	**94.58**	100.00	**98.38**	**97.61**	**97.38**	**94.93**	**97.76**
BERT	GA	100.00	65.21	65.11	72.86	69.72	72.60	100.00	77.46	71.10	76.52	76.43	80.27	100.00	75.82	70.64	72.66	73.81	79.94
	PWWS	100.00	60.48	64.37	66.93	70.04	75.95	100.00	75.94	69.80	70.33	75.98	79.20	100.00	78.79	74.28	75.13	74.11	82.66
	DPSO	100.00	69.12	66.07	75.63	70.85	77.23	100.00	80.71	74.54	79.60	82.36	81.01	100.00	83.10	75.97	79.00	78.81	84.58
	DQPSO	100.00	69.77	66.22	76.52	70.73	77.45	100.00	81.24	74.62	78.51	82.32	81.23	100.00	82.82	76.14	78.90	77.95	84.44
	HIQPSO	100.00	69.14	66.11	76.65	71.12	77.87	100.00	80.38	74.61	78.39	82.67	81.17	100.00	81.94	76.09	77.10	77.56	84.44
	TextFooler	100.00	67.25	64.22	75.09	68.98	74.59	100.00	83.80	74.76	79.38	78.53	82.83	100.00	83.78	79.14	80.27	78.61	86.10
	CLARE	100.00	68.48	64.64	72.55	69.29	75.67	100.00	82.84	76.22	77.18	78.28	84.00	100.00	73.03	70.70	70.97	70.63	81.84
	DynHI	100.00	**99.88**	**99.81**	**99.32**	**99.20**	**98.52**	100.00	**98.09**	**97.70**	**88.84**	**87.13**	**96.41**	100.00	**97.21**	**94.50**	**92.13**	**91.02**	**97.06**

Table 7
Attack success rates (%) of different methods during real-time attacks on the BERT model updating process on SNLI.

Attack Method	2-E %s at i		4-E %s at i				6-E %s at i						8-E %s at i								3111%s at i			
	1	2	1	2	3	4	1	2	3	4	5	6	1	2	3	4	5	6	7	8	1	2	3	4
PWWS	64.1	61.9	63.2	66.1	65.5	62.5	60.8	57.1	60.9	58.8	54.2	57.0	62.3	58.9	63.9	58.9	62.9	60.1	66.3	64.8	59.5	63.9	62.2	66.2
DPSO	79.3	81.4	**84.6**	82.4	74.7	82.6	77.3	83.0	79.5	80.9	70.7	80.7	78.0	73.4	73.5	75.2	80.9	79.4	**84.3**	80.3	77.8	78.1	79.8	79.3
HIQPSO	81.4	81.5	84.2	**83.6**	76.7	**85.0**	77.3	**85.2**	81.0	**82.9**	73.3	**82.5**	78.0	75.1	76.2	77.1	82.2	**81.8**	**84.3**	**84.0**	78.5	79.8	**82.6**	**82.5**
TextFooler	**82.3**	**83.0**	81.9	80.8	**80.1**	81.3	**82.6**	82.0	**81.6**	**82.9**	**80.0**	82.0	**79.3**	**79.7**	**81.2**	**80.0**	**82.3**	80.4	82.6	82.6	**81.0**	**81.2**	81.0	82.1
CLARE	24.1	23.5	24.4	25.7	23.5	23.6	25.4	23.5	23.3	25.3	19.1	24.7	26.9	22.3	23.9	22.5	25.5	24.2	26.0	26.1	22.4	34.3	34.4	36.6

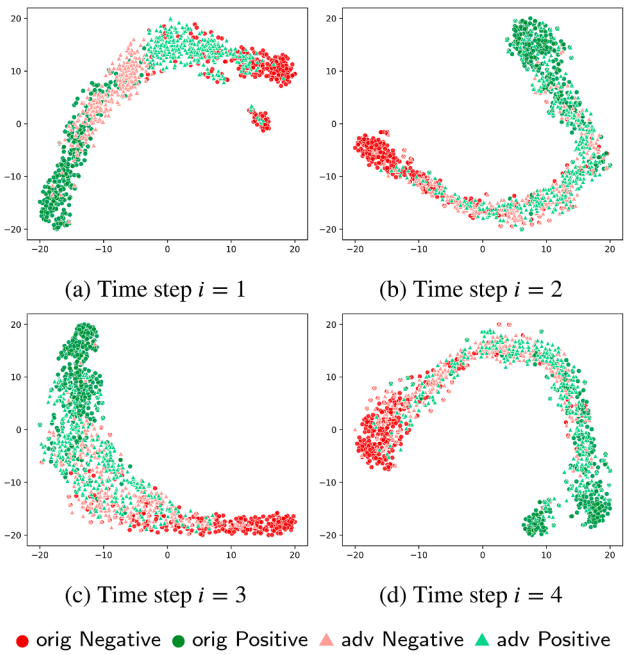


Fig. 8. Visualization of prediction distribution of BERT on IMDB under 4-E split. The adversarial samples are generated using HIQPSO method.

4.8. Visualization of prediction distributions

To verify the impact of prediction distribution changes in continuously updating models on static adversarial samples, we used the BERT model to predict adversarial samples in $\boldsymbol{X}^{\text{Valid}}$ and their corresponding original samples. We extracted the output vector of the [CLS] token as a sentence embedding and used the t-SNE algorithm to visualize the results, as shown in Fig. 8. At the initial time step, the negative and positive adversarial samples clustered on the positive and negative sides of the decision boundary, respectively, forming two distinct groups. However, as the model continued training in subsequent time steps, these adversarial samples no longer formed clear clusters but instead became irregularly distributed. This transition from distinct clusters to a disordered distribution visually demonstrates that the prediction distribution of the model has shifted. The adversarial samples lost their effectiveness because the attack path, which relies on the decision boundary of the old model, was no longer suitable for the prediction distributions of the new models trained with the subsequent data.

Furthermore, in subsequent time steps, although the adversarial samples no longer formed two distinct clusters, their distribution was still primarily located in the intermediate region between the positive and negative original sample clusters. This suggests that although the prediction distribution of the model has shifted from the initial time step, it still retains certain similarities to the distribution at that time step. Otherwise, many adversarial samples would have fallen into the clusters of the original samples.

4.9. Real-time attack during model updating

4.9.1. Real-time attack using static methods

To evaluate the effectiveness of generating adversarial samples in real time during the model updating process, we performed attacks on $\mathcal{M}_i$ at each time step, with the results presented in Table 7. For each adversarial method, we generated adversarial samples from the same original samples at every time step under the same splitting scheme to ensure comparability. It can be observed that the attack performance of different methods at later time steps is comparable to that at the initial time step, indicating that continuous model updates do not exhibit a significant improvement in robustness against adversarial attacks.

The experimental methods and metrics proposed in this paper are specifically designed for realistic scenarios where frequent real-time adversarial generation is infeasible. The results in Table 7 show that, when sufficient attack budget is available, the existing adversarial methods can maintain better effectiveness by performing real-time attacks. Therefore, these findings also demonstrate that the proposed experimental methods and metrics can systematically explain the failure patterns of the existing static adversarial methods when they attack the continuously updating models under constrained attack budgets.

4.9.2. Potential dynamic real-time attack strategy

Execution of real-time attacks can indeed maintain strong attack effectiveness of the existing static adversarial methods. However, it also incurs a higher time cost, since these existing methods regenerate adversarial samples from the original ones at each time step, with the samples generated in previous time steps being discarded. To improve the attack effectiveness of adversarial methods in the continuously updating model scenarios, two targeted approaches in two different situations can be pursued. The first is that, under tight resource constraints, it is necessary to improve the performance of static adversarial methods themselves. The second is that, with a moderate amount of resources, adversarial methods can be upgraded from “static” to “dynamic”, enabling dynamic real-time attacks that better adapt to the continuously updating model scenarios.

To improve the efficiency of dynamic real-time attacks, based on the results of dynamic attacks by using static adversarial methods, adversarial samples for later time steps should be derived from those generated at earlier steps. When $\mathcal{M}_i$ is attacked, attacks are attempted on samples in $\boldsymbol{X}^{\text{Valid}}$. If sample $X'_j \in \boldsymbol{X}^{\text{Valid}}$ succeeds, it is reused as the adversarial sample for the current time step; otherwise a new adversarial sample is generated from X'_j rather than from the original sample. In addition, samples from previous time steps may be aggregated, selectively fused according to potential specific strategies, and used to produce new adversarial samples. However, since this work focuses on examining how the effectiveness of current static adversarial methods changes in continuously updating model scenarios, detailed discussion of specific dynamic real-time attack strategies is left beyond the scope of this paper.

4.10. Further discussion

This paper systematically investigates text adversarial attacks in continuously updating model scenarios, and the experiments are based on commonly used classification models to accurately reflect the performance changes of existing methods. However, we should acknowledge that this study does not include experiments on Large Language Models (LLMs). Since LLMs are generative models with different inference mechanisms, attacking them would require additional methodological developments, such as redesigning fine-tuning strategies and attack pipelines. Consequently, this is beyond the scope of the current study. Future work will investigate adversarial attacks targeting LLMs and explore how to extend the proposed framework to the LLM setting.

5. Conclusion

This paper presented a new adversarial attack task that focuses on studying whether static text adversarial methods in continuously updating modeling scenarios can maintain their original performance. To standardize the research in this field, we proposed a comprehensive evaluation framework that explicitly decouples attack performance into two distinct capabilities. This framework integrates two experimental methods, three evaluation metrics, and a lightweight dynamic attack baseline. Experimental results demonstrated that continuously updating models lead to a significant performance degradation of static adversarial samples, whereas the dynamic baseline maintained high effective-

ness. This contrast confirms that the proposed evaluation framework can effectively measure the performance and limitations of adversarial methods in continuously updating model scenarios. In particular, a higher single dynamic attack success rate indicates that the generated samples have stronger direct effectiveness against individual model updates, a higher continuous dynamic attack success rate reflects that the generated samples have superior iterative persistence throughout the continuous updating process, and a higher overall dynamic attack success rate suggests that the generated samples have more comprehensive capability in balancing these two capabilities and can cause a greater damage to the continuously updating model. Our future work will focus on developing adversarial methods that have better attack performance for the dynamic attack problem proposed in this work.

CRediT authorship contribution statement

Zihao Yu: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft; **Jun Sun:** Writing – review & editing, Supervision, Resources; **Qidong Chen:** Formal analysis, Writing – original draft; **Vasile Palade:** Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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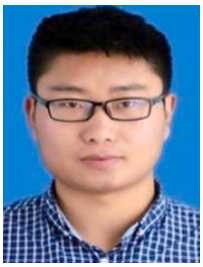

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